Reviewer report: pahaz/sshtunnel#304: Migrate to PEP518

Run `sshtunnel-304`, tested 2026-10-21 19:49 IST by ops/daily_run.py (route source-pair). Proof-of-Change commit `5492834eef22efd122549356c73d26b3d0ae040a`. Report generated 2026-10-21 19:50 IST by reports/pr_report.py.

Baseline: **full** (test comparison + claim probes + coverage gate + claim flagging (differential probes shown, informational)).

COULD NOT VERIFY

Recommended action: **Re-run** Confidence: **Low**

The run did not produce a usable comparison, so there is no conclusion about the PR. The run did not finish cleanly, so it says nothing about the PR.

Advice only: Proof-of-Change never blocks a merge and never posts on GitHub.

Summary in plain English

- **What it says it does:** Migrate to PEP518 (pahaz/sshtunnel#304).
- **What we observed:** the run failed its own checks: 3 alternating runs per side.
- **What you should do:** Do not decide yet. Fix the problem under 'How to fix' and run the check again.

{ "summary": \"The PR \"Migrate to PEP518\" received a \"needs_attention\" verdict. The probe reported \"probe: 8 observation(s) need attention: meta.author differs and no claim covers it; meta.author_email differs and no claim covers it; meta.download_url differs and no claim covers it; meta.home_page differs and no claim covers it;\", plus the reviewer-installed-metadata probe has no measured line coverage and no tests could be mapped to the touched files.\"}

Quoted from the record: `Migrate to PEP518`; `needs_attention`; `probe: 8 observation(s) need attention: meta.author differs and no claim covers it; meta.author_email differs and no claim covers it; meta.download_url differs and no claim covers it; meta.home_page differs and no claim covers it;`

Model summary is advisory; the code verdict above is authoritative.

Agent swarm (advisory)

Role	Model	Status	Notes
judge	nvidia/Nemotron-3-Ultra-550b-a55b	ok	1 labels
helper	nvidia/Nemotron-3_5-Lightning	ok	meta.author: no PEP518 claim, ignore in future diffs; meta.author_email: no PEP518 claim, ignore in future diffs; meta.download_url: pre-existing, ignore in PEP518 diffs
summary	nvidia/nemotron-3-super-120b-a12b	ok	-

judge/helper/summary are advisory; code owns the verdict

Demo helper

These notes are background from the console and the daily check. They do not change the verdict.

If setup fails, the helper retries the install up to 2 times. If a test still can't be verified because something is missing, it installs what's missing and reruns that test once.

- Setup check: GitHub token: present.
- Setup check: Nebius token: present.
- Setup check: free memory 6.3 GB (of 15.6 GB).
- Setup check: free disk 98.1 GB.
- Demo helper: git is installed.
- Demo helper: python3 is installed.
- Demo helper: uv is installed.
- Looked up with Tavily: 4 results
- Query: sshtunnel python setup install instructions https://github.com/pahaz/sshtunnel site:github.com/pahaz/sshtunnel
- Query: sshtunnel python documentation install https://github.com/pahaz/sshtunnel site:github.com/pahaz/sshtunnel
- Query: sshtunnel python install requirements https://pypi.org/project/sshtunnel/ https://github.com/pahaz/sshtunnel site:pypi.org
- Source: sshtunnel · PyPI — https://pypi.org/project/sshtunnel/
- Source: GitHub - pahaz/sshtunnel: SSH tunnels to remote server. · GitHub — https://github.com/pahaz/sshtunnel
- Source: sshtunnel/sshtunnel.py at master · pahaz/sshtunnel · GitHub — https://github.com/pahaz/sshtunnel/blob/master/sshtunnel.py
- Source: psycopg2 conn.poll() hangs when used with sshtunnel · Issue #135 · pahaz/sshtunnel · GitHub — https://github.com/pahaz/sshtunnel/issues/135
- Setup plan: the same packages on main and pr.
- Setup plan: paramiko is noted, installed with the project.
- Setup plan: install docutils (listed in requirements.txt) on main and pr.
- Setup plan: install sphinx (listed in requirements.txt) on main and pr.
- Setup plan: install sphinxcontrib-napoleon (listed in requirements.txt) on main and pr.
- Setup plan: install bashtest (listed in requirements-syntax.txt) on main and pr.
- Setup plan: install check-manifest (listed in requirements-syntax.txt) on main and pr.
- Setup plan: install flake8 (listed in requirements-syntax.txt) on main and pr.
- Setup plan: install mccabe (listed in requirements-syntax.txt) on main and pr.
- Setup plan: install pygments (listed in requirements-syntax.txt) on main and pr.
- Setup plan: install readme (listed in requirements-syntax.txt) on main and pr.
- Setup plan: install twine (listed in requirements-syntax.txt) on main and pr.
- Setup plan: install coveralls (listed in requirements.txt) on main and pr.
- Setup plan: install mock (listed in requirements.txt) on main and pr.
- Setup plan: install pytest (listed in requirements.txt) on main and pr.
- Setup plan: install pytest-cov (listed in requirements.txt) on main and pr.
- Setup plan: install pytest-xdist (listed in requirements.txt) on main and pr.
- Setup plan: install bleach<5.0.0 (listed in requirements.txt) on main and pr.
- Setup plan: install readme-renderer<25.0; python_version == '3.4' (listed in requirements.txt) on main and pr.
- Setup plan: install importlib-metadata==1.7.0; python_version == '3.5' (listed in requirements.txt) on main and pr.
- Setup plan: install tox>=1.8.1 (listed in the test extra) on main and pr.
- Setup plan: Python 3.12: Python 3.12 already meets >=2.7

- Installed docutils 0.22.4 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed sphinx 9.1.1 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed sphinxcontrib-napoleon 0.7 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed bashtest 0.0.8 from requirements-syntax.txt (listed in requirements-syntax.txt) on main and pr.
- Installed check-manifest 0.51 from requirements-syntax.txt (listed in requirements-syntax.txt) on main and pr.
- Installed flake8 7.4.1 from requirements-syntax.txt (listed in requirements-syntax.txt) on main and pr.
- Installed mccabe 0.7.0 from requirements-syntax.txt (listed in requirements-syntax.txt) on main and pr.
- Installed pygments 2.21.0 from requirements-syntax.txt (listed in requirements-syntax.txt) on main and pr.
- Installed readme 0.7.1 from requirements-syntax.txt (listed in requirements-syntax.txt) on main and pr.
- Installed twine 7.0.0 from requirements-syntax.txt (listed in requirements-syntax.txt) on main and pr.
- Installed coveralls 4.1.0 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed mock 5.2.0 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed pytest 9.1.1 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed pytest-cov 7.1.0 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed pytest-xdist 3.8.0 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed bleach 4.1.0 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed readme-renderer 46.0 from requirements.txt (listed in requirements.txt) on main and pr.
- Installed importlib-metadata from requirements.txt (listed in requirements.txt) on main and pr.
- Installed tox 4.67.1 from pyproject (listed in the test extra) on main and pr.

Background (related public bug reports, not used for the verdict)

1. "Issues · pahaz/sshtunnel · GitHub" https://github.com/pahaz/sshtunnel/issues "pahaz/sshtunnel#294 · opened on Nov 6, 2024 ### Transfer the local port to remote server pahaz/sshtunnel#293 · opened on Oct 29, 2024 ### Using sshtunnel on Mac? pahaz/sshtunnel#290 · opened on Mar 20, 2024 1 ### SSH Tunnel Connection Drop..." 2. "Welcome to sshtunnel's documentation! — sshtunnel 0.4.0 documentation" https://sshtunnel.readthedocs.io "v.0.0.4.1 (Pahaz) : + fix CLI issues #13 (Pahaz) v.0.0.4 (Pahaz) : + daemon mode by default for all threads (JM Fernández, Pahaz) - incompatible + move `make_ssh_forward_server` to `SSHTunnelForwarder.make_ssh_forward_server` (Pahaz, JM Fer..." 3. "python-sshtunnel(1) — Arch manual pages" https://man.archlinux.org/man/python-sshtunnel.1.en "daemon mode by default for all threads (JM Fernández , Pahaz) - incompatible move make_ssh_forward_s...

Claims vs reality

No tests ran, so no claim could be checked.

What changed in behavior

Not measured: probe: 8 observation(s) need attention: meta.author differs and no claim covers it; meta.author_email differs and no claim covers it; meta.download_url differs and no claim covers it; meta.home_page differs and no claim covers it; probe reviewer-installed-metadata has no measured line coverage (no changed non-test Python files); it does not count as covering the change; tests: no tests could be mapped to the touched files.

Probes

Model-written probes: model provider: nebius (Token Factory, live).

A probe is a small deterministic script run on both sides in the same environments as the tests, 3 alternating runs per side. It prints JSON observations; plain code (poc/diff.py) compares them. A difference that no claim of the PR covers needs attention. Every probe was frozen (content sha256) before any PR-side code ran, and checked against that freeze before each PR-side run. Only a live model probe is blind; fixed probes and recorded model fixtures are known-answer cases.

reviewer-installed-metadata (fixed (post-hoc, written after results were known), `probes/pahaz/sshtunnel/304.py`, frozen `24ce09957e0a` at 2026-10-21T14:19:50.217257+00:00): Installed package metadata, console script and module import before vs after moving setup.py into pyproject.toml.

- P-C1 (PR title (Migrate to PEP518)): Moving the build config to pyproject.toml keeps what users install the same: name, version, runtime dependency, extras, console script, classifiers and importable module. Expects the covered observations to stay the same.

Observation	Claim	Result	What the probe saw (main -> PR)	Log lines (main, PR)
`meta.author`	-	differs, not in the PR description	`value`: "Pahaz White" -> null	probe-reviewer-installed-metadata-main-1.jsonl:1 «"observation": "meta.author"» probe-reviewer-installed-metadata-pr-1.jsonl:1 «"observation": "meta.author"»
`meta.author_email`	-	differs, not in the PR description	`value`: "pahaz.white@gmail.com" -> "Pahaz White <pahaz.white@gmail.com>"	probe-reviewer-installed-metadata-main-1.jsonl:2 «"observation": "meta.author_email"» probe-reviewer-installed-metadata-pr-1.jsonl:2 «"observation": "meta.author_email"»
`meta.classifiers`	P-C1	as claimed	same on both sides: `["Development Status :: 3 - Alpha", "Intended Audience :: Developers", "License :: OSI ...`	probe-reviewer-installed-metadata-main-1.jsonl:3 «"observation": "meta.classifiers"» probe-reviewer-installed-metadata-pr-1.jsonl:3 «"observation": "meta.classifiers"»
`meta.console_scripts`	P-C1	as claimed	same on both sides: `["sshtunnel=sshtunnel:_cli_main"]`	probe-reviewer-installed-metadata-main-1.jsonl:4 «"observation": "meta.console_scripts"» probe-reviewer-installed-metadata-pr-1.jsonl:4 «"observation": "meta.console_scripts"»
`meta.description_length_nonzero`	-	same on both sides	same on both sides: `true`	probe-reviewer-installed-metadata-main-1.jsonl:5 «"observation": "meta.description_length_nonzero"» probe-reviewer-installed-metadata-pr-1.jsonl:5 «"observation": "meta.description_length_nonzero"»

Observation	Claim	Result	What the probe saw (main -> PR)	Log lines (main, PR)
`meta.download_url`	-	differs, not in the PR description	`value` : "https://pypi.python.org/packages/sou... -> null	probe-reviewer-installed-metadata-main-1.jsonl:6 «"observation": "meta.download_url"» probe-reviewer-installed-metadata-pr-1.jsonl:6 «"observation": "meta.download_url"»
`meta.home_page`	-	differs, not in the PR description	`value` : "https://github.com/pahaz/sshtunnel" -> null	probe-reviewer-installed-metadata-main-1.jsonl:7 «"observation": "meta.home_page"» probe-reviewer-installed-metadata-pr-1.jsonl:7 «"observation": "meta.home_page"»
`meta.keywords`	-	differs, not in the PR description	`value` : "ssh tunnel paramiko proxy tcp-forward" -> "ssh,tunnel,paramiko,proxy,tcp-forward"	probe-reviewer-installed-metadata-main-1.jsonl:8 «"observation": "meta.keywords"» probe-reviewer-installed-metadata-pr-1.jsonl:8 «"observation": "meta.keywords"»
`meta.license`	-	same on both sides	same on both sides: `"MIT"`	probe-reviewer-installed-metadata-main-1.jsonl:9 «"observation": "meta.license"» probe-reviewer-installed-metadata-pr-1.jsonl:9 «"observation": "meta.license"»
`meta.name`	P-C1	as claimed	same on both sides: `"sshtunnel"`	probe-reviewer-installed-metadata-main-1.jsonl:10 «"observation": "meta.name"» probe-reviewer-installed-metadata-pr-1.jsonl:10 «"observation": "meta.name"»
`meta.platform`	-	differs, not in the PR description	`value` : ["macos", "unix", "windows"] -> []	probe-reviewer-installed-metadata-main-1.jsonl:11 «"observation": "meta.platform"» probe-reviewer-installed-metadata-pr-1.jsonl:11 «"observation": "meta.platform"»
`meta.project_urls`	-	differs, not in the PR description	`value` : [] -> ["Download, https://pypi.python.org/p...	probe-reviewer-installed-metadata-main-1.jsonl:12 «"observation": "meta.project_urls"» probe-reviewer-installed-metadata-pr-1.jsonl:12 «"observation": "meta.project_urls"»
`meta.provides_extra`	P-C1	as claimed	same on both sides: `["build-sphinx", "dev", "test"]`	probe-reviewer-installed-metadata-main-1.jsonl:13 «"observation": "meta.provides_extra"» probe-reviewer-installed-metadata-pr-1.jsonl:13 «"observation": "meta.provides_extra"»

Observation	Claim	Result	What the probe saw (main -> PR)	Log lines (main, PR)
`meta.requires_dist`	P-C1	as claimed	same on both sides: `["check-manifest; extra == \"dev\"", "paramiko>=2.7.2", "sphinx; extra == \"build-sphin...`	probe-reviewer-installed-metadata-main-1.jsonl:14 «"observation": "meta.requires_dist"» probe-reviewer-installed-metadata-pr-1.jsonl:14 «"observation": "meta.requires_dist"»
`meta.requires_python`	-	differs, not in the PR description	`value` : null -> ">=2.7"	probe-reviewer-installed-metadata-main-1.jsonl:15 «"observation": "meta.requires_python"» probe-reviewer-installed-metadata-pr-1.jsonl:15 «"observation": "meta.requires_python"»
`meta.summary`	P-C1	as claimed	same on both sides: `"Pure python SSH tunnels"`	probe-reviewer-installed-metadata-main-1.jsonl:16 «"observation": "meta.summary"» probe-reviewer-installed-metadata-pr-1.jsonl:16 «"observation": "meta.summary"»
`meta.version`	P-C1	as claimed	same on both sides: `"0.4.0"`	probe-reviewer-installed-metadata-main-1.jsonl:17 «"observation": "meta.version"» probe-reviewer-installed-metadata-pr-1.jsonl:17 «"observation": "meta.version"»
`module.has_cli_main`	P-C1	as claimed	same on both sides: `true`	probe-reviewer-installed-metadata-main-1.jsonl:18 «"observation": "module.has_cli_main"» probe-reviewer-installed-metadata-pr-1.jsonl:18 «"observation": "module.has_cli_main"»
`module.has_forwarder`	P-C1	as claimed	same on both sides: `true`	probe-reviewer-installed-metadata-main-1.jsonl:19 «"observation": "module.has_forwarder"» probe-reviewer-installed-metadata-pr-1.jsonl:19 «"observation": "module.has_forwarder"»
`module.version`	P-C1	as claimed	same on both sides: `"0.4.0"`	probe-reviewer-installed-metadata-main-1.jsonl:20 «"observation": "module.version"» probe-reviewer-installed-metadata-pr-1.jsonl:20 «"observation": "module.version"»

Citation check (PASS): 40 citation(s), every one re-read from its log file and matched.

Was the changed code exercised?

Coverage: not needed (no changed Python lines).

Coverage gate (no-executable-change, confidence normal): no executable code changed (docs, tests or config only), so there was nothing to cover.

How this is measured: one extra, untimed run of the PR side's tests under coverage.py. Changed lines come from comparing the tested main and PR files. Compiled code (C, C++, Cython) is invisible to coverage.py; such a change only counts as verified when the PR adds or changes a test for that module.

What went right

- Tested `6ca35a37971a` of the PR against main (source-pair route).
- Changed code was exercised (coverage gate): no executable code changed (docs, tests or config only), so there was nothing to cover.
- Probes ran on both sides: 1 probe(s), 3 alternating runs per side, every run printed its observations.
- Single-thread environment: OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1, BLIS_NUM_THREADS=1, PYTHONHASHSEED=0.
- Machine type recorded: local x86_64 8 cpus.
- Nothing posted upstream: GitHub was only read.
- Every probe observation cites its output line: 40 citation(s), every one re-read from its log file and matched.

What went wrong and how to fix it

Issue	Status	Detail	How to fix
3 alternating runs per side	FAIL	no test runs recorded	Re-run; see setup logs.

Evals and guardrails

Check	Result	Detail
3 alternating runs per side	FAIL	no test runs recorded
Changed code was exercised (coverage gate)	PASS	no executable code changed (docs, tests or config only), so there was nothing to cover
Probes ran on both sides	PASS	1 probe(s), 3 alternating runs per side, every run printed its observations
Single-thread environment	PASS	OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1, BLIS_NUM_THREADS=1, PYTHONHASHSEED=0
Machine type recorded	PASS	local x86_64 8 cpus
Nothing posted upstream	PASS	GitHub was only read
Known-answer replay (fresh re-run)	SKIP	not replayed yet

Check	Result	Detail
Every claim cites a real log line	SKIP	no log lines to cite (no tests ran)
Every probe observation cites its output line	PASS	40 citation(s), every one re-read from its log file and matched

Checklist

- ☐ I read the verdict and the plain-English summary.
- ☐ Every 'false' or 'unmentioned change' row has a log line I can open.
- ☐ Nothing under 'What went wrong' is a run problem (those mean Re-run, not Reject).
- ☐ I recorded my decision (see below).

Decision recorded: none yet. To record one, run `make decide REPORT=/report/sshtunnel-304 DECISION=approve` (or `reject`, `approve_with_notes`, `re_run`).

Rerun command

```
make check PR=https://github.com/pahaz/sshtunnel/pull/304
```

Evidence links

- Pull request pahaz/sshtunnel#304
- record.json
- pr.diff
- setup-main.log
- setup-pr.log
- git.log
- probe-reviewer-installed-metadata-main-1.jsonl
- probe-reviewer-installed-metadata-pr-1.jsonl

Glossary

Term	Meaning
main	The library as it is today, without the PR.
PR	Pull request: a proposed change to the library.
run	One full pass of the selected tests. We do 3 per side, alternating main and PR.
flaky	A test that gives different results when nothing changed; it is ignored as evidence.
log line	A line in a pytest-*.log file: the raw output the verdict is based on.

Term	Meaning
route	How the two sides were built: nightly+patch, source-build or source-pair (see docs/OPERATIONS.md).
changed lines covered	How many of the PR's changed, executable Python lines ran during the PR side's tests.
re-parametrized	A test whose parameter ids changed (renamed cases); the new ids are judged on their own.